



Combinatorics 2

- Raghav Goel

Goal

To understand:

- Stars and Bars
- Independent Choices

Previous class:-

- Binomial Coefficient

- Optimal Implementation ^{nC_r}
(Template)

- Arrangements



Stars and Bars

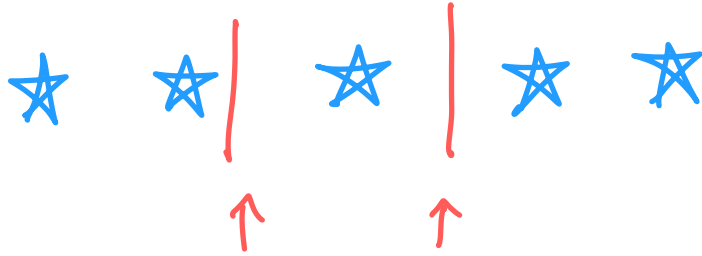


Number of ways to distribute N identical objects into K distinct boxes

[https://en.wikipedia.org/wiki/Stars_and_bars_\(combinatorics\)](https://en.wikipedia.org/wiki/Stars_and_bars_(combinatorics)) ✓

$$\binom{n + k - 1}{k - 1}$$

5 stars



2 bars



all ☆ are similar

all | are similar

5 objects + 2 new objects

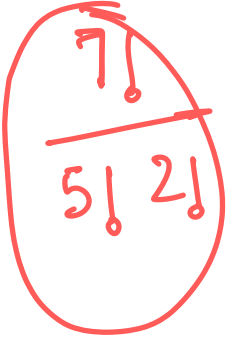
7 objects

$7C_2$

7



$7C_2$



N stars

K boxes

4 boxes

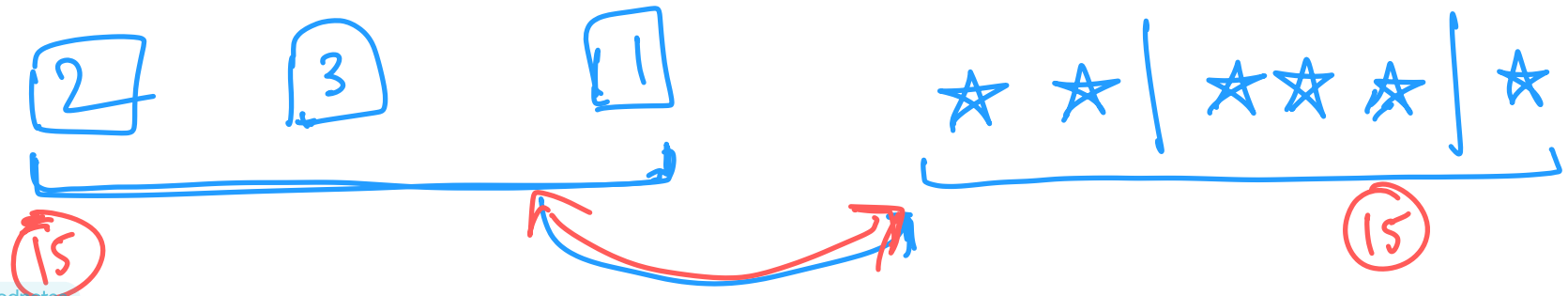
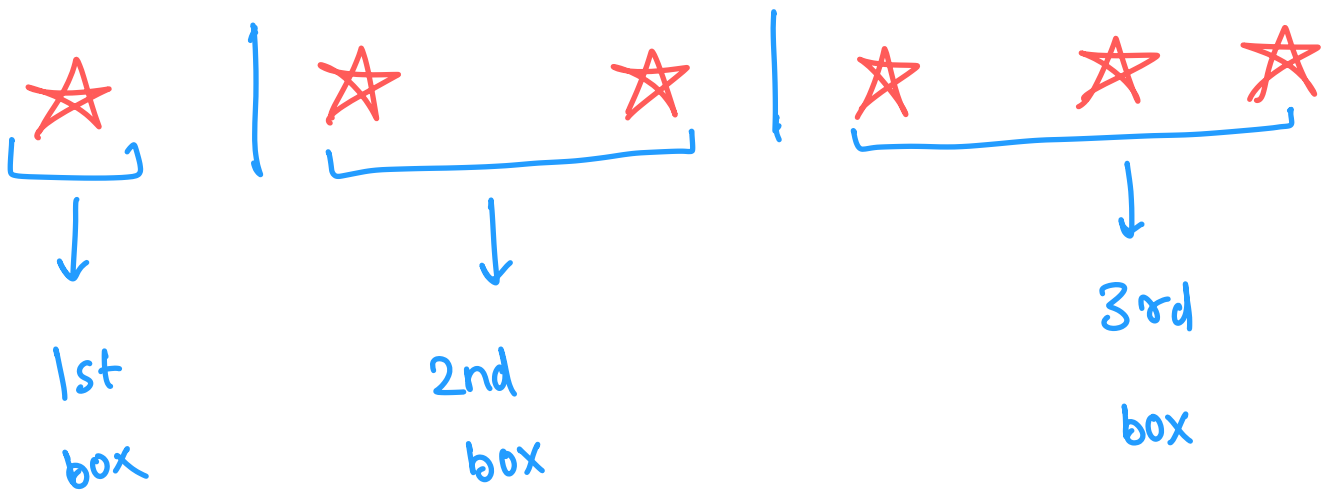
10 ★

10	0	0	0
4	3	2	1
9	1	0	0

different from
→

0 10 0 0

6 stars
3 boxes



One to one relation

for each arrangement in "^{similar} Stars into ^{distinct} boxes",
there is exactly one arrangement of
putting "^{similar} bars among the stars"
similar

and vice versa.

similar
 N stars

distinct
 K boxes

N identical stars and $(K-1)$ identical bars

$$\frac{(N+K-1)!}{N! (K-1)!}$$

$$\frac{(N+K-1)!}{(K-1)!}$$

or $\frac{(N+K-1)!}{N!}$

$\frac{(N+K-1)!}{(K-1)!}$

N identical
 K distinct

$K-1$
 $\downarrow$
distinct

N stars

$(K-1)$ bars

2nd proof

1 bar $\rightarrow N+1$ option
2 bar $\rightarrow N+2$
3 bar $\rightarrow N+3$
 $\vdots$
 $K-1$ bar $\rightarrow N+K-1$

|| ★ ★ ★

| ★ | ★ ★

$$\frac{(N+K-1)!}{N! (K-1)!}$$

$N + K - 1$

N are identical

$K-1$ identical

$\underbrace{N}_{aaaa} \underbrace{K-1}_{bbbb}$

$$\frac{(N+K-1)!}{N! (K-1)!} \leftarrow \frac{(cnt_a + cnt_b)!}{cnt_a! cnt_b!}$$

Example

$$10+3-1 \text{ } {}_{3-1}C_{3-1} = {}_{12}C_2$$



- You have 10 identical candies and want to distribute them among 3 children. Each child can receive any number of candies, including zero.
- A manager has 7 identical tasks to assign to 4 employees. Each employee should handle at least 1 task.

$$7-4$$

4 employees

$$3+4-1 \text{ } {}_{4-1}C_{4-1} = {}_6C_3$$

3 identical task left

no. of ways to distribute these tasks such that atleast one employee doesn't get any task

total ways - (ways where everyone gets atleast one task)

7 tasks



4 employee

≥ 1 task

give 1 task
each

3 tasks

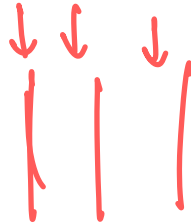


4 employees

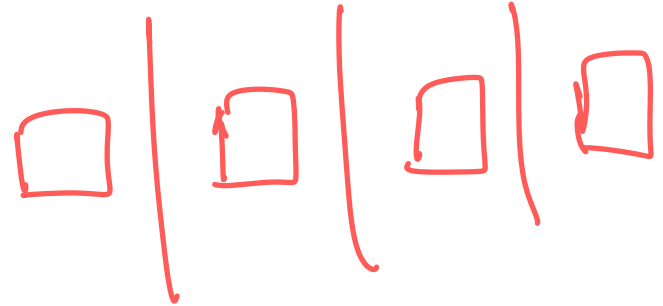
≥ 0 tasks

$$\boxed{n-1 \atop k-1}$$

$t \quad t \quad t$



3t



$$\boxed{6 \atop 3}$$



Stars and Bars Applications

What are the number of ways to solve the equation below?

$x_1 + x_2 + x_3 = 6 \Rightarrow x_1=1, x_2=2, x_3=3$

$$\underline{x_1 + x_2 + \dots + x_k = n}$$

$x_i \geq 0$

K boxes
 n identical objects

$n+k-1 \choose k-1$



Stars and Bars Applications

What are the number of ways to solve the equation below?

$$x_1 + x_2 + \cdots + x_k = n$$

$$x_i \geq 1$$

$$x_1 + x_2 + \dots + x_k = n$$

$$x_i \geq 1$$

$$(x_1' + 1) + (x_2' + 1) + \dots + (x_k' + 1) = n$$

$$x_1' + x_2' + \dots + x_k' = \boxed{n - k}$$

$$x_i' = x_i - 1$$

$$\boxed{x_i = x_i' + 1}$$

$$x_i' \geq 0$$

$$x_i' \geq 0$$

$$(n - k) + k - 1$$

$$C_{k-1}$$

$\Rightarrow$

$$\boxed{n - 1 \atop C_{k-1}}$$



Stars and Bars Applications

What are the number of ways to solve the equation below?

$$x_1 + x_2 + \cdots + x_k = n$$

$$x_i \geq a_i$$

$$x_i \geq a_i$$

$$x'_i = x_i - a_i$$

$$x_i = x'_i + a_i$$

$$x_i - a_i \geq 0$$



$$x'_i \geq 0$$

$$a = \{1, 2, 3\}$$

$$x_1 + x_2 + \dots + x_k = n$$

$$(x'_1 + a_1) + (x'_2 + a_2) + \dots + (x'_k + a_k) = n$$

$$x'_1 + x'_2 + \dots + x'_k = n - \sum a_i$$

$$(n - \sum a_i + k - 1)$$

$$C_{k-1}$$



Problem: Distributing Apples

distinct

identical

There are n children and m apples that will be distributed to them. Your task is to count the number of ways this can be done. ($1 \leq n, m \leq 10^6$)

Since the answer might be too huge, print it modulo $10^9 + 7$.

Problem Link - <https://cses.fi/problemset/task/1716>

$$\frac{n+m-1}{n-1} C_{n-1}$$



Problem: Array

Chris the Rabbit has been interested in arrays ever since he was a child. At the moment he is researching arrays with the length of n , containing only integers from 1 to n . He is not good at math, that's why some simple things drive him crazy. For example, yesterday he grew keen on counting how many different beautiful arrays there are. Chris thinks that an array is beautiful if it meets one of the two conditions:

- each elements, starting from the second one, is no more than the preceding one
- each element, starting from the second one, is no less than the preceding one

→ *non-increasing*
→ *non-decreasing*

Having got absolutely mad at himself and at math, Chris came to Stewie and Brian to ask them for help. However, they only laughed at him and said that the answer is too simple and not interesting. Help Chris the Rabbit to find the answer at last.

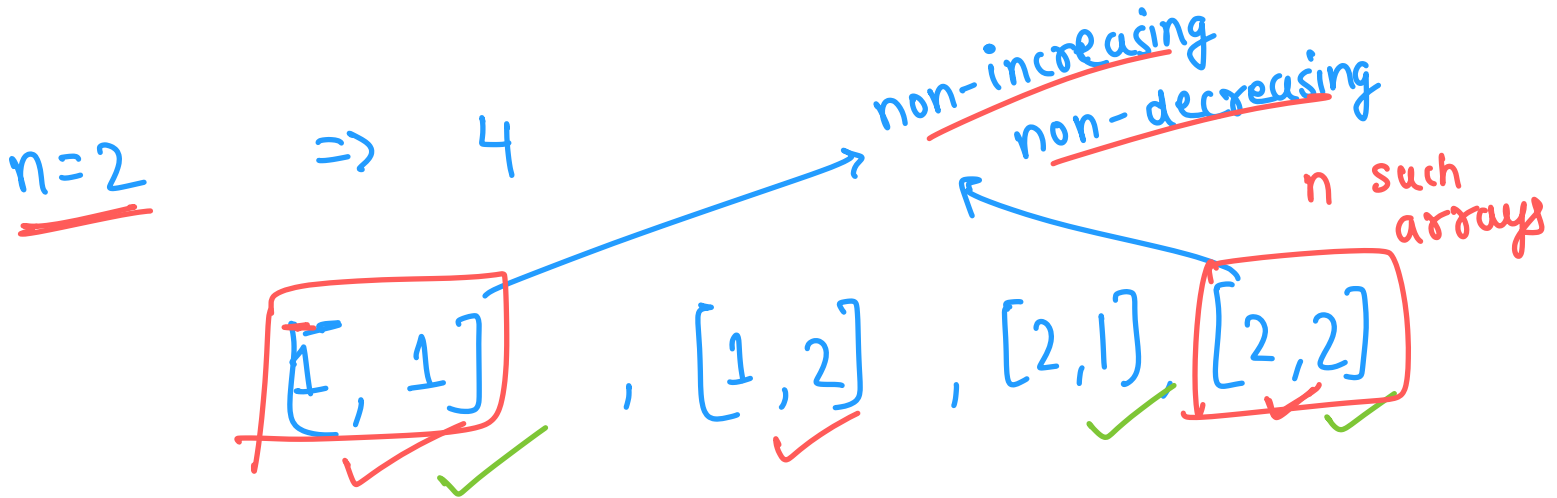
Input

The single line contains an integer n which is the size of the array ($1 \leq n \leq 10^5$).

Problem Link - <https://codeforces.com/contest/57/problem/C>

$$n \leq 10^5$$

no. of non-increasing or non-decreasing arrays



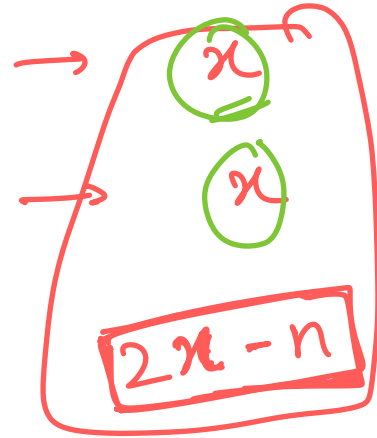
number of non-decreasing arrays

non-increasing arrays

$[1, 2, 2, 3]$

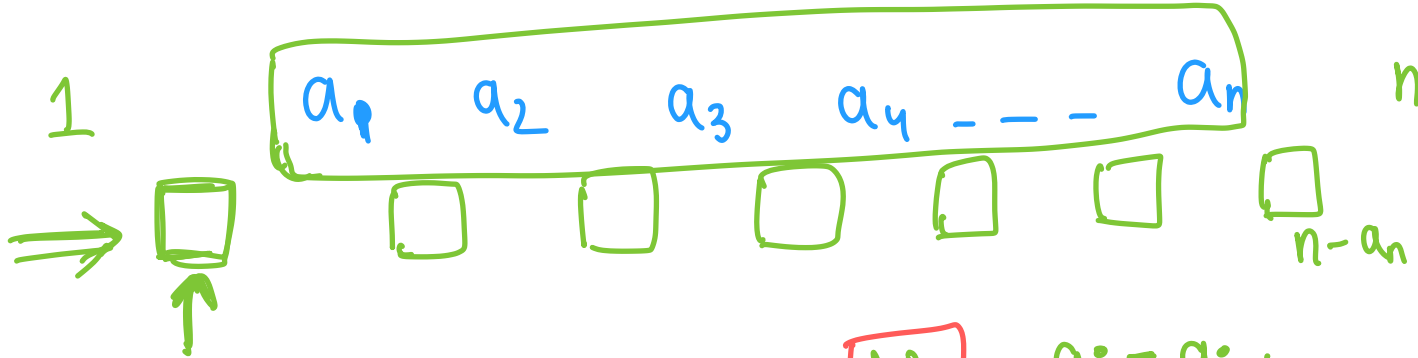
$[n, n-1, n-1, n-2]$

$$a \rightarrow (n - a_i + 1)$$



$a_i - 1$

1900



$$b_i = a_i - a_{i-1}$$

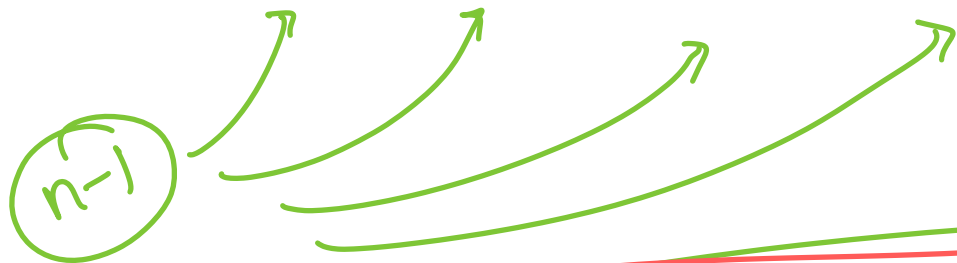
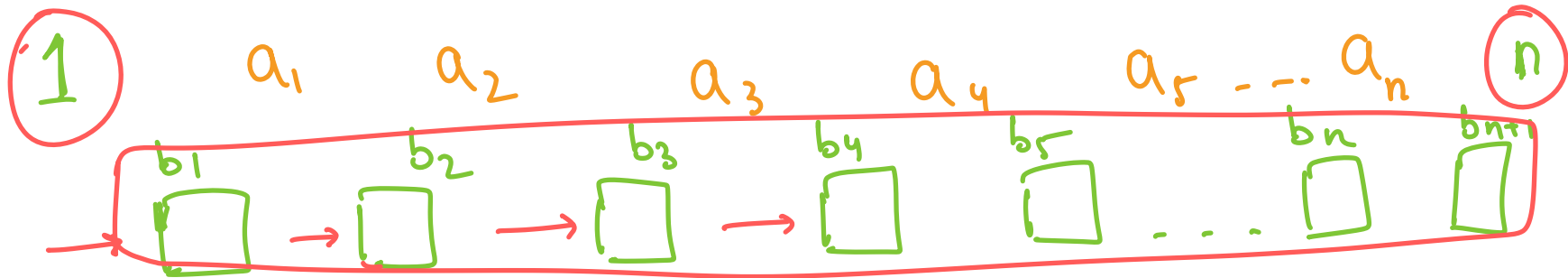
$$\sum_{i=1}^{n+1} b_i = n-1 \rightarrow b_1 + b_2 + \dots + b_k = N \rightarrow \binom{N+k-1}{k-1}$$

$$K = n+1$$

$$N = n-1$$

$$1 \quad 1 \quad , \quad 1 \quad , \quad 1 \quad , \quad 1 \quad 4$$

$$\boxed{0} + \boxed{0} + \boxed{0} + \boxed{0} + \boxed{3} = n-1$$



$$b_1 + b_2 + \dots + b_{n+1} = n-1$$

$$(n-1) + (n+1) - 1$$

$$C_{(n+1)-1} \Rightarrow$$

$$\binom{2n-1}{n}$$

$$n=2$$

$${}^3C_2 \Rightarrow \textcircled{3}$$

$$x=3$$

$$2x-n \Rightarrow 4$$

$$n=3$$

$${}^{2n-1}C_n \Rightarrow {}^5C_3 \Rightarrow 10$$

$$x=10$$

$$2x-n = 17$$

$$2 * \binom{2n-1}{n} = n$$

$$\begin{array}{ccccc}
 x & & + & & x & = & 2x \\
 \text{non-decreasing} & & & & \text{non-increasing} \\
 \checkmark & & & & \checkmark
 \end{array}$$

$\{ [1, 1, 1, 1, 1], [2, 2, \dots, 2], [3, \dots, 3], \dots, [n, \dots, n] \}$

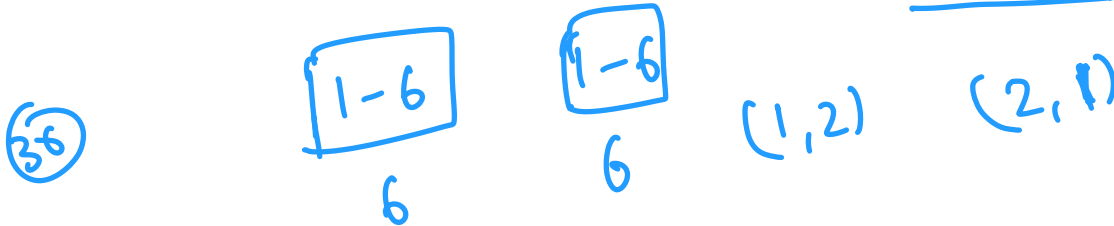
final ans $\rightarrow 2x - n$



Independent Choices

- Independent choices refer to **scenarios where decisions or selections made in one step do not influence the options available in other steps.**
- When choices are independent, the total number of outcomes is calculated by **multiplying the number of options available at each step.**

Example: Total number of outcomes on rolling 2 dice simultaneously.



	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

36 different outcomes

K distinct dice
find the total no. of possible outcomes

die $\rightarrow$ 1 to 6

$$6^k$$

$$\underline{6} * \underline{6} * \underline{6} * \underline{6} * \dots * \underline{6}$$



Key Formula for Independent Choices

If there are n independent steps, and each step i has k_i choices, the total number of outcomes is:

$$\text{Total Outcomes} = k_1 * k_2 * k_3 * \dots * k_n$$

Example → Tossing a coin



Problem: Lucky Numbers

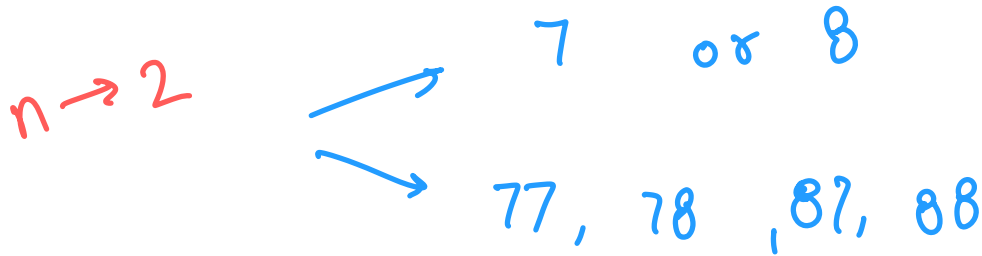
The numbers of all offices in the new building of the Tax Office of IT City will have lucky numbers.

Lucky number is a number that consists of digits 7 and 8 only. Find the maximum number of offices in the new building of the Tax Office given that a door-plate can hold a number not longer than n digits.

Problem Link - <https://codeforces.com/problemset/problem/630/C>

find the total number of numbers such that :-

- the number consists of only 7 or 8
- the number of digits $\leq n$



$\swarrow$ 1 digit $\rightarrow$ $\underline{2} \rightarrow 2$
 $\swarrow$ 2 digits $\rightarrow$ $\underline{2} * \underline{2} \rightarrow 4$
 $\swarrow$ 3 digits $\rightarrow$ $\underline{2} \times \underline{2} \times \underline{2} \rightarrow 8$
 $\swarrow$ i digits $\rightarrow$ $\underline{2} \ \underline{2} \ \underline{2} \ \underline{2} \ \underline{2} \dots \underline{2} \rightarrow 2^i$

$$\underline{\underline{2^1 + 2^2 + 2^3 + \dots + 2^n}}$$

$$\boxed{2^{n+1} - 1 - 1}$$

$$= \cancel{2^1} + 2^1 + 2^2 + \dots + 2^n$$

$$\boxed{2^{n+1} - 2}$$



Problem: Anagrams

You are given a string `s` containing one or more words. Every consecutive pair of words is separated by a single space ' '.

$$|s| \leq 10^5$$

A string `t` is an **anagram** of string `s` if the i^{th} word of `t` is a **permutation** of the i^{th} word of `s`.

- For example, "acb dfe" is an anagram of "abc def", but "def cab" and "adc bef" are not.

Return the number of **distinct anagrams** of `s`. Since the answer may be very large, return it **modulo** $10^9 + 7$.

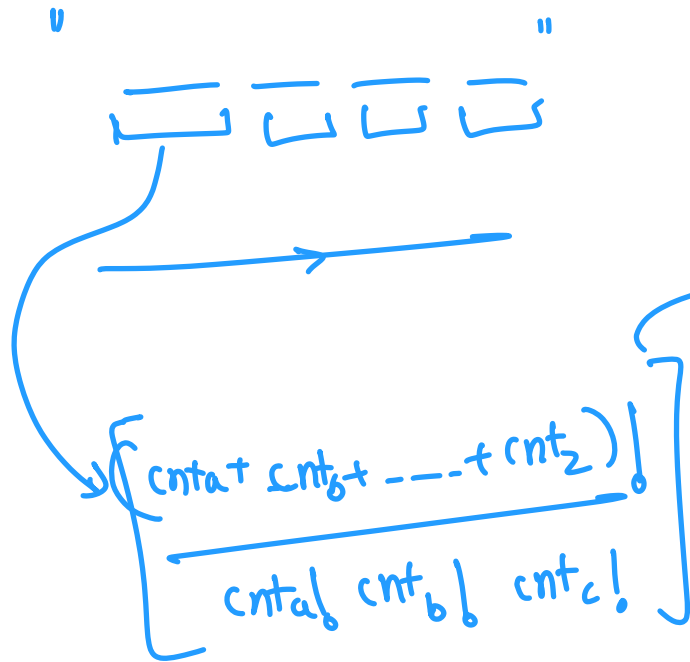
Problem Link - <https://leetcode.com/problems/count-anagrams>

"abccd" →

number of distinct permutations
of "abccd"

$$\begin{array}{l} s = \text{"abc def ghi abb"} \\ \quad \downarrow 6 \quad \downarrow 6 \quad \downarrow 6 \quad \downarrow 3 \\ t = \underline{\hspace{1cm}} \quad \underline{\hspace{1cm}} \quad \underline{\hspace{1cm}} \quad \underline{\hspace{1cm}} \end{array} \rightarrow \begin{array}{l} 6 \\ \text{unique} \\ \text{permutations} \\ \text{of } \underline{\underline{abc}} \end{array}$$

$6 * 6 * 6 * 3$



$$ans = ans \times \left[\frac{(cnt_a + cnt_b + \dots + cnt_z)!}{cnt_a! cnt_b! cnt_c!} \right] \% MOD$$



Bonus Problems

- Monotonic Remuneration -
<https://codeforces.com/problemset/problem/1102/E>
- The Intriguing Obsession -
<https://codeforces.com/problemset/problem/869/C>
- Right Triangles -
<https://leetcode.com/problems/right-triangles/description/>
- Count Ways to Make Array With Product -
<https://leetcode.com/problems/count-ways-to-make-array-with-product/description/>